

LECTURES 13,14

GRAPHS – DIRECTED, UNDIRECTED, WEIGHTED

Directed graph $G = (V, E)$

where V is the set of vertices of G , and E is a set of ordered pairs such as (a, b) representing **DIRECTED** edges between pairs of vertices. E is a relation on V .

As a data structure, a graph can be represented in the the form of an **adjacency list** or an **adjacency matrix**.

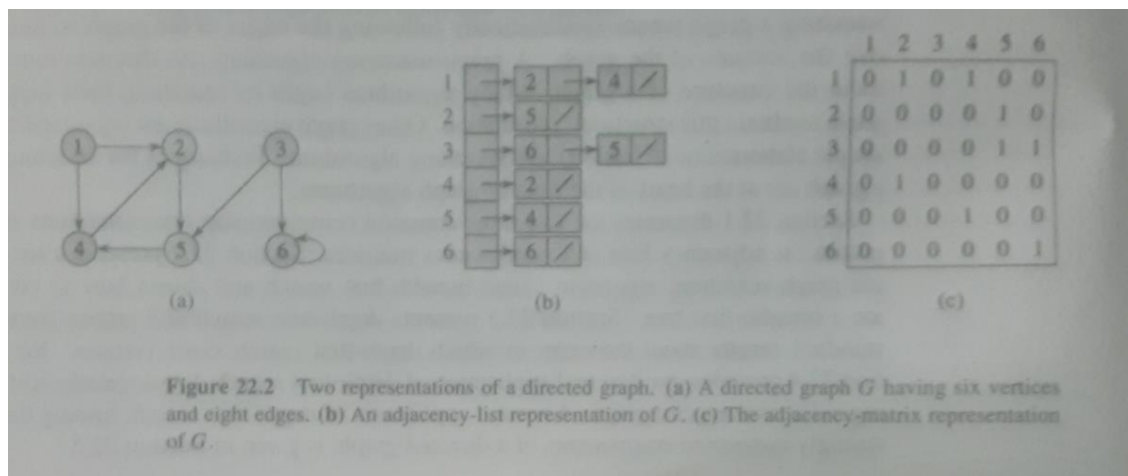


Figure 22.2 Two representations of a directed graph. (a) A directed graph G having six vertices and eight edges. (b) An adjacency-list representation of G . (c) The adjacency-matrix representation of G .

Undirected graph $G = (V, E)$

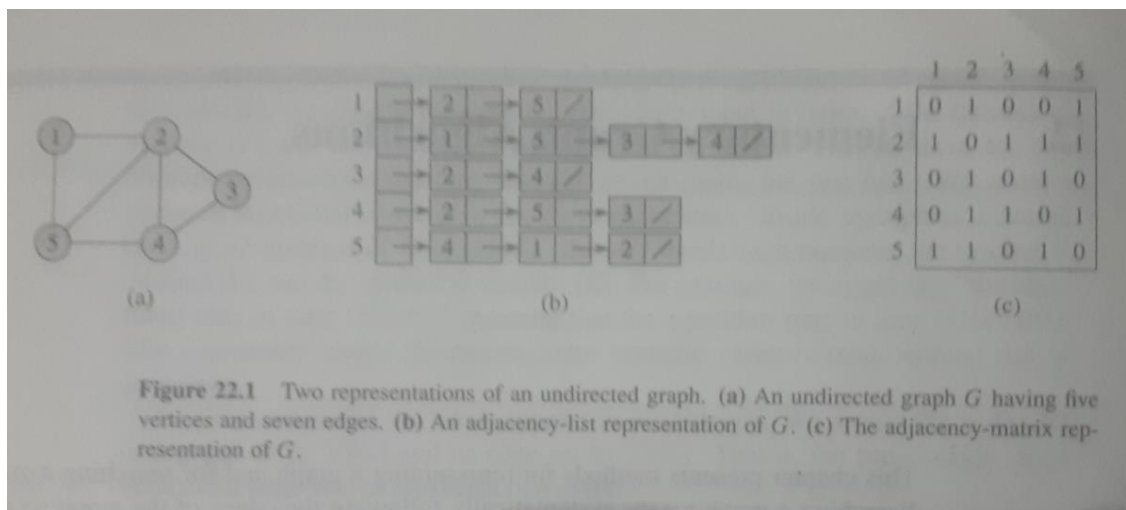


Figure 22.1 Two representations of an undirected graph. (a) An undirected graph G having five vertices and seven edges. (b) An adjacency-list representation of G . (c) The adjacency-matrix representation of G .

Every edge is now in the form $\{a,b\}$. Note (1) the size of the adjacency list, and (2) symmetry of the adjacency matrix.

Weighted graphs

A **weight** is associated with every edge – typically representing a cost.

Huge number of practical applications.

Recall “routing algorithms” of various kinds.

Questions:

1. What is the longest possible path in a graph which does not “loop back on itself” anywhere?
2. If A is the adjacency matrix of a graph G , what does $A \cdot A = A^2$ represent? What do $A^3 \dots A^k$ represent?
3. How can we generate a (weighted) graph with, say, 1000 nodes?
4. How can we **graphically** represent a (weighted) graph?

PROPERTIES OF A BINARY RELATION – that is, a directed graph

Reflexivity

Transitivity

Symmetry

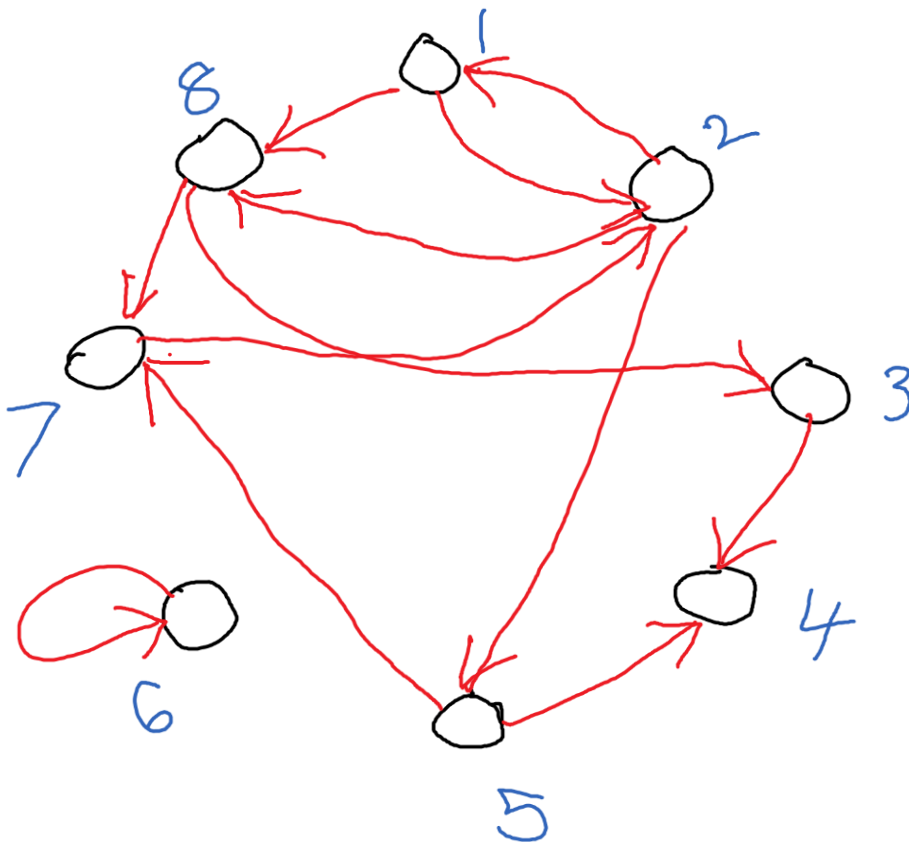
Anti-symmetry

How to generate a graph using the function rand()?

See the accompanying Excel sheet.

0	1	0	0	0	0	0	1
1	0	0	0	1	0	0	1
0	0	0	1	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	1	0
0	0	0	0	0	1	0	0
0	1	0	0	0	0	0	0
0	0	1	0	0	0	1	0

Adjacency matrix (randomly generated) of a graph with $|V| = 8$.



Visual representation of the same graph. Note that $|E| = 12$.

A **path** can be constructed in a graph by following the edges from one vertex to the next. Thus, assuming that the respective edges exist in the graph, the sequence of vertices $v_1, v_2 \dots v_n$ define a path of length $n-1$.

A **cycle** is a path in which the initial and final vertices are the same.

UNLESS STATED EXPLICITLY OTHERWISE – in the context of the algorithms we study – no **vertex** or **edge** will be repeated in a path or a cycle.

A directed graph is **strongly connected** if any vertex u is **reachable** from any other vertex v .

If a graph is NOT strongly connected, then it must have two or more **strongly connected components**.

Simple exercise: Let us look for the presence or absence of these properties in the above graph.

Question: Let A be the adjacency matrix of a graph G . What is the significance of the following matrix?

$$\bigcup_{k=1, n-1} A^k$$

→ Meaning and significance of **transitive closure**, **reachability**.

Database representation of adjacency list -- PREPARATION FOR LAB

Vertex table has one record per vertex; assume that the primary key is vertex number.

Edge table has one record per edge; compound key is (source vertex, destination vertex).

In practice, vertices may represent cities, and then edges may represent road/ rail/ flight connectivity. Edge weights can be added as, say, travel times or costs; in practice, other properties would be required.

Graph can be represented as a relational database. Functions are available such as create table, add record, find record ... *et cetera*.

Note: Such a graph does NOT yet represent geographical data.